

04/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

PST-5

Time : 180 Mins.

Answers

1. (4)	37. (1)	73. (4)	109. (4)	145. (3)
2. (3)	38. (4)	74. (4)	110. (2)	146. (1)
3. (3)	39. (2)	75. (1)	111. (3)	147. (4)
4. (4)	40. (1)	76. (3)	112. (2)	148. (2)
5. (1)	41. (3)	77. (3)	113. (1)	149. (3)
6. (3)	42. (2)	78. (2)	114. (3)	150. (2)
7. (3)	43. (3)	79. (3)	115. (3)	151. (4)
8. (2)	44. (4)	80. (3)	116. (3)	152. (4)
9. (1)	45. (2)	81. (2)	117. (2)	153. (4)
10. (4)	46. (4)	82. (1)	118. (3)	154. (2)
11. (2)	47. (4)	83. (4)	119. (1)	155. (4)
12. (3)	48. (2)	84. (2)	120. (4)	156. (2)
13. (4)	49. (3)	85. (3)	121. (3)	157. (4)
14. (2)	50. (1)	86. (1)	122. (1)	158. (1)
15. (1)	51. (4)	87. (1)	123. (2)	159. (1)
16. (1)	52. (2)	88. (1)	124. (1)	160. (4)
17. (2)	53. (4)	89. (4)	125. (2)	161. (4)
18. (2)	54. (3)	90. (4)	126. (4)	162. (1)
19. (3)	55. (1)	91. (2)	127. (2)	163. (1)
20. (2)	56. (2)	92. (1)	128. (3)	164. (3)
21. (1)	57. (4)	93. (3)	129. (3)	165. (2)
22. (1)	58. (3)	94. (1)	130. (3)	166. (4)
23. (1)	59. (4)	95. (4)	131. (1)	167. (3)
24. (4)	60. (2)	96. (3)	132. (4)	168. (3)
25. (2)	61. (1)	97. (1)	133. (2)	169. (2)
26. (3)	62. (3)	98. (2)	134. (2)	170. (1)
27. (4)	63. (4)	99. (1)	135. (2)	171. (4)
28. (3)	64. (3)	100. (4)	136. (1)	172. (2)
29. (2)	65. (1)	101. (3)	137. (4)	173. (3)
30. (2)	66. (2)	102. (3)	138. (2)	174. (2)
31. (3)	67. (4)	103. (2)	139. (3)	175. (3)
32. (3)	68. (3)	104. (1)	140. (1)	176. (2)
33. (1)	69. (3)	105. (1)	141. (1)	177. (1)
34. (2)	70. (2)	106. (1)	142. (3)	178. (3)
35. (3)	71. (3)	107. (1)	143. (4)	179. (3)
36. (2)	72. (3)	108. (3)	144. (2)	180. (3)

Hints and Solutions

PHYSICS

(1) Answer : (4)

Solution:

Heat required if 250 g of ice melts completely, $Q_1 = mL$

$$Q_1 = 250 \times 80 = 20,000 \text{ cal} \dots (i)$$

Heat released if 600 g of water reaches 0°C , $Q_2 = ms\Delta T$

$$Q_2 = 600 \times 1 \times 18 = 10,800 \text{ cal} \dots (ii)$$

Since heat required is greater than heat released, hence system will contain some ice at equilibrium.

Amount of ice that melts to become water

$$Q = mL \Rightarrow 10800 = m \times 80$$

$$m = 135 \text{ g}$$

$$\therefore \text{Ice left} = 250 - 135 = 115 \text{ g}$$

(2) Answer : (3)

Solution:

Black body will have more rate of emission as well as absorption

(3) Answer : (3)

Solution:

Expansion of glass container is same as liquid, therefore volume expansion coefficients of both of them are equal.

$$\gamma_{\lambda} = \gamma_g$$

$$\Rightarrow \gamma = 3\alpha$$

(4) Answer : (4)

Solution:

In reversible isochoric process, change in volume is zero and hence work done is zero.

(5) Answer : (1)

Solution:

$$\text{Efficiency} = 1 - \frac{T_L}{T_H} = 1 - \frac{Q'}{Q}$$

$$\therefore \frac{Q'}{Q} = \frac{T}{T+4}$$

$$\Rightarrow Q' = \frac{Q}{4}$$

(6) Answer : (3)

Solution:

Slope of adiabatic curve is more than isothermal at a given state. Clearly process B to C have more slope than A to B , hence A to B is isothermal whereas B to C is adiabatic.

(7) Answer : (3)

Solution:

Temperature can never go below 0 K , and hence it cannot be negative.

Molecular motion ceases at 0 K .

(8) Answer : (2)

Solution:

When the metallic ball is heated, then all the linear dimensions will increase according to

$$\Delta L = L_0 \alpha \Delta T$$

$$\text{For area; } A = 4\pi R^2$$

$$\frac{\Delta A}{A} = \frac{2\Delta R}{R}$$

$$\text{For volume; } V = \frac{4}{3}\pi R^3$$

$$\frac{\Delta V}{V} = \frac{3\Delta R}{R}$$

Clearly more percentage increase will be for volume.

(9) Answer : (1)

Solution:

$$Q = ms\Delta T$$

Heat gained by liquid A = Heat lost by liquid B

$$m \times S_A (38 - 30) = m \times S_B (40 - 38)$$

$$S_A \times 8 = S_B \times 2$$

$$\Rightarrow \frac{S_A}{S_B} = \frac{1}{4}$$

(10) Answer : (4)

Solution:

$$Q = KA \frac{\Delta T}{\Delta x}$$

$$80 = K\pi r^2 \times \frac{\Delta T}{l}$$

$$Q' = K \times \pi (2r)^2 \times \frac{\Delta T}{2l} = 2 \left(K\pi r^2 \frac{\Delta T}{l} \right) = 160 \text{ cal/s}$$

(11) Answer : (2)

Solution:

Total kinetic energy of a gas depends on its temperature.

Gas molecules move in random direction with different velocities.

(12) Answer : (3)

Solution:

Particles of medium execute SHM when a transverse progressive sinusoidal wave passes through it, and these particles vibrate with different phase.

(13) Answer : (4)

Solution:

Total energy of a particle remains constant.

$$\therefore E \propto \chi^2$$

(14) Answer : (2)

Solution:

Time period of a simple pendulum is given by

$$T = 2\pi \sqrt{\frac{l}{g}}$$

On the surface of moon acceleration due to gravity is $\frac{g}{6}$.

$$\therefore 2\pi \sqrt{\frac{l}{g}} = 2\pi \sqrt{\frac{l}{g/6}}$$

$$\Rightarrow \frac{1}{g} = \frac{6l}{g}$$

$$\Rightarrow l = \frac{1}{6} \text{ m}$$

(15) Answer : (1)

Solution:

Time period of spring-mass system

$$T = 2\pi \sqrt{\frac{m}{K_{\text{eff}}}}$$

For series combination of springs

$$K_{\text{eff}} = \frac{K_1 K_2}{K_1 + K_2}$$

$$\therefore T = 2\pi \sqrt{\frac{(K_1 + K_2)m}{K_1 K_2}}$$

(16) Answer : (1)

Solution:

$$W = P\Delta V = 50(4 - 8) = -200 \text{ J}$$

$$\text{Given } Q = 120 \text{ J}$$

$$Q = \Delta U + W$$

$$120 = \Delta U - 200$$

$$\Rightarrow \Delta U = 320 \text{ J}$$

(17) Answer : (2)

Solution:

In isothermal process, $\Delta U = 0$.

Therefore, work done on the gas will result in release of heat.

(18) Answer : (2)

Solution:

$$v = \omega \sqrt{A^2 - x^2}$$

Given $A = 8$ cm, $\omega = 5$ rad/s, $x = 2$ cm

$$v = 5 \sqrt{8^2 - 2^2} = 5 \sqrt{60} \text{ cm/s}$$

(19) Answer : (3)

Solution:

$$F = -\frac{dU}{dx} = -\frac{d}{dx} [-U_0 \cos(\alpha x)]$$

$$F = U_0(-\sin \alpha x)\alpha$$

$$F = -U_0 \alpha \sin \alpha x$$

If $\alpha \rightarrow 0$, then $\sin \alpha x \rightarrow \alpha x$

Also for SHM, $F = -kx$

$$\therefore -U_0 \alpha \times \alpha x = -kx$$

$$k = U_0 \alpha^2$$

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{m}{U_0 \alpha^2}} \Rightarrow T = \frac{2\pi}{\alpha} \sqrt{\frac{m}{U_0}}$$

(20) Answer : (2)

Solution:

$$(C_V)_{mix} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2}$$

$$(C_V)_{mix} = \frac{1 \times \frac{3}{2}R + 1 \times \frac{5}{2}R}{1+1}$$

$$(C_V)_{mix} = 2R$$

(21) Answer : (1)

Solution:

Amplitude of SHM for equation $y = \sin \omega t + B$ is equal to A .

(22) Answer : (1)

Solution:

$$v_{avg} = \sqrt{\frac{8RT}{\pi M}}$$

$$\Rightarrow v \propto \sqrt{T}$$

$$\frac{v}{v'} = \sqrt{\frac{300}{T'}}$$

$$\Rightarrow \frac{v}{2v} = \sqrt{\frac{300}{T}}$$

$$T = 1200 \text{ K}$$

(23) Answer : (1)

Solution:

$$\int dQ = \int nC dT$$

$$Q = \int 2\alpha T dT$$

$$= 2\alpha \int_{T_0}^{2T_0} T dT$$

$$= 2\alpha \frac{T^2}{2} \Big|_{T_0}^{2T_0}$$

$$Q = 3\alpha T_0^2$$

(24) Answer : (4)

Solution:

Kinetic energy of an ideal gas is a function of temperature (in kelvin), hence the graph is straight line starting from zero kelvin.

(25) Answer : (2)

Solution:

The ball will repeat its motion after regular interval of time hence motion is periodic, but its acceleration is not varying as $a = -\omega^2 x$ hence its not SHM.

(26) Answer : (3)

Solution:

$$\text{Given } x = (0.03\text{m}) \sin\left(50t + \frac{\pi}{3}\right)$$

$$v = \frac{dx}{dt} = (0.03) \times 50 \cos\left(50t + \frac{\pi}{3}\right)$$

$$v_{\max} = 0.03 \times 50 = \frac{3}{2} \text{ m/s}$$

$$(\text{Kinetic energy})_{\max} = \frac{1}{2} m \times v_{\max}^2$$

$$= \frac{1}{2} \times 2 \times \left(\frac{3}{2}\right)^2 = \frac{9}{4} \text{ J}$$

(27) Answer : (4)

Solution:

$$y = A \sin\left(\frac{2\pi}{\lambda} bt - \frac{2\pi}{\lambda} x\right)$$

Comparing with $y = A \sin(\omega t - kx)$, we get

$$\omega = \frac{2\pi b}{\lambda} = 2\pi f$$

$$f = \frac{b}{\lambda}$$

(28) Answer : (3)

Solution:

The collisions amongst the molecules as well as with the walls are perfectly elastic in nature.

(29) Answer : (2)

Solution:

For ideal gas : $PV = nRT$

Put $n = 1$, we get $\frac{PV}{T} = R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

(30) Answer : (2)

Solution:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\therefore \frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}}$$

(31) Answer : (3)

Solution:

A rigid diatomic molecule has 3 translational and 2 rotational degrees of freedom.

(32) Answer : (3)

Solution:

Beat frequency, $\Delta f = 234 - 230 = 4 \text{ Hz}$

Time interval between two successive maxima or minima is, $t = \frac{1}{\Delta f}$

$$t = \frac{1}{4} = 0.25 \text{ s}$$

(33) Answer : (1)

Solution:

$$y = 0.2 \sin(5x) \cos(300t)$$

$$K = \frac{2\pi}{\lambda} = 5$$

$$\Rightarrow \lambda = \frac{2\pi}{5} \text{ m}$$

Distance between two successive nodes is $\frac{\lambda}{2}$.

$$\text{Hence } \Delta x = \frac{2\pi}{5} \times \frac{1}{2} = \frac{\pi}{5} \text{ m}$$

(34) Answer : (2)

Solution:

For maxima of sound at D , path difference = $n\lambda$

Path difference = $3\lambda - \lambda = 2\lambda$

$$\therefore 2\lambda = n\lambda$$

$$\lambda = \frac{2\lambda}{n}$$

For $\lambda_{\max}$, $n = 1$

$$\therefore \lambda = 2\lambda$$

(35) Answer : (3)

Solution:

Velocity of sound in air

$$V = \sqrt{\frac{\gamma RT}{M}}$$

$$\Rightarrow V \propto \sqrt{T}$$

(36) Answer : (2)

Solution:

Work done in isothermal process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

Work done in adiabatic process

$$W = \frac{P_i V_i - P_f V_f}{\gamma - 1}$$

Change in internal energy

$$\Delta U = nC_V \Delta T$$

Heat exchange in isobaric process

$$Q = nC_p \Delta T$$

(37) Answer : (1)

Solution:

Newton's law of cooling :

$$\frac{\Delta T}{\Delta t} = K \left(\frac{T_1 + T_2}{2} - T_s \right)$$

$$\frac{90^\circ - 80^\circ}{5} = K \left(\frac{90^\circ + 80^\circ}{2} - 20^\circ \right)$$

$$2 = K(65) \dots (i)$$

$$\frac{80^\circ - 70^\circ}{t} = K (75^\circ - 20^\circ)$$

$$\frac{10}{t} = \frac{2}{65} \times 55$$

$$t = \frac{10 \times 65}{2 \times 55} \simeq 6 \text{ minutes}$$

(38) Answer : (4)

Solution:

Acceleration of particle in SHM

$$a = -\omega^2 x$$

$$\omega = \frac{2\pi}{T} \Rightarrow \omega = \frac{2\pi}{4} = \frac{\pi}{2} \text{ rad/s}$$

$$a = -\left(\frac{\pi}{2}\right)^2 \times 0 = 0$$

(39) Answer : (2)

Solution:

2nd overtone of a closed organ pipe:

$$f_1 = \frac{5v}{4L_C} \dots (i)$$

3rd overtone of an open organ pipe:

$$f_2 = \frac{4v}{2L_0} \dots (ii)$$

$$\text{Given } f_1 = f_2$$

$$\therefore \frac{5v}{4L_C} = \frac{4v}{2L_0}$$

$$\Rightarrow \frac{L_C}{L_0} = \frac{5}{8}$$

(40) Answer : (1)

Solution:

Speed of transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}} \dots (i)$$

μ will remain constant and tension is quadrupled, then

$$v' = \sqrt{\frac{4T}{\mu}} = 2 \sqrt{\frac{T}{\mu}} \dots (ii)$$

$$\frac{v}{v'} = \frac{1}{2}$$

(41) Answer : (3)

Solution:

 Given $A = 0.4 \text{ m}$, $v = 180 \text{ m s}^{-1}$, $\lambda = 60 \text{ m}$

$$y = A \sin(\omega t - kx + \phi_0)$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{60} = \frac{\pi}{30}$$

$$v = \frac{\omega}{k} \Rightarrow \omega = vk = 180 \times \frac{\pi}{30} = 6\pi$$

$$\therefore y = 0.4 \sin\left(6\pi t - \frac{\pi}{30}x + \phi_0\right)$$

(42) Answer : (2)
Solution:

Wien's displacement law:

$$\lambda T = b$$

$$10^{-6} T_1 = 2 \times 10^{-6} \times T_2$$

$$\Rightarrow \frac{T_1}{T_2} = 2$$

(43) Answer : (3)
Solution:

 The given process is adiabatic, so $Q = 0$ but the gas does work in expansion.

$$\text{Hence, } 0 = \Delta U + W$$

Temperature of gas will decrease due to decrease in internal energy.

(44) Answer : (4)
Solution:

 For an ideal gas, $PV = nRT$

In isothermal process, temperature remains constant.

$$P\Delta V + V\Delta P = 0 \Rightarrow P\Delta V = -V\Delta P$$

$$\frac{\Delta P}{P} = -\frac{\Delta V}{V}$$

(45) Answer : (2)
Solution:

 For adiabatic compression $PV^\gamma = \text{constant}$

$$PV_0^{5/3} = P' \times \left(\frac{V_0}{8}\right)^{5/3}$$

$$\Rightarrow P' = P \times (2)^5$$

$$P' = 32P$$

CHEMISTRY

(46) Answer : (4)
Solution:

For isoelectronic species, ionic size of cation decreases as the charge on cation increases. For anion, ionic size increases as the charge on anion increases.

Ion Size (pm)

$$\text{K}^+ - 138$$

$$\text{Ca}^{2+} - 100$$

$$\text{Al}^{3+} - 53.5$$

(47) Answer : (4)
Solution:

 In a subshell, the maximum number of orbitals is $(2\ell + 1)$ and each orbital has 2 electrons. Hence, maximum number of electrons in a subshell $= 2(2\ell + 1) = 4\ell + 2$ electrons.

(48) Answer : (2)
Solution:

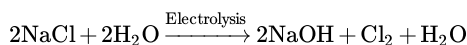
	Percentage	Moles	Simplest ratio
C	38.71	$\frac{38.71}{12} = 3.22$	$\frac{3.22}{3.02} \approx 1$
H	9.67	$\frac{9.67}{1} = 9.67$	$\frac{9.67}{3.02} \approx 3$

$$\left| \begin{array}{c|c|c} \text{O} & 48.38 & \frac{48.38}{16} = 3.02 \\ \hline & & \frac{3.02}{3.02} \approx 1 \end{array} \right|$$

Empirical formula $\Rightarrow \text{CH}_3\text{O}$

(49) Answer : (3)

Solution:



2 mol of NaCl $\equiv$ 1 mol of Cl_2

$$\therefore \frac{1.17}{58.5} \text{ mol of NaCl} \equiv \frac{1}{2} \times \frac{1.17}{58.5} \Rightarrow 0.01 \text{ mol}$$

At STP, volume of 1 mol of $\text{Cl}_2 = 22.4 \text{ L}$

Hence, volume of 0.01 mol of $\text{Cl}_2 = 22.4 \times 0.01$

0.224 L

(50) Answer : (1)

Solution:

$$1. \text{ Oxygen atoms in } \text{CO}_2 = \frac{4.48}{22.4} \times N_A \times 2 \Rightarrow 0.4 N_A$$

$$2. \text{ Oxygen atoms in } \text{H}_2\text{O} = \frac{5.4}{18} \times N_A \Rightarrow 0.3 N_A$$

$$3. \text{ Oxygen atoms in CO} = \frac{5.6}{28} \times N_A = 0.2 N_A$$

$$4. \text{ Oxygen atom in SO}_2 = 0.1 \times N_A \times 2 = 0.2 N_A$$

(51) Answer : (4)

Solution:

1 molal aqueous solution means 1 mole solute present in 1 kg of solvent

$$\text{So, mole fraction of solute} = \frac{1}{1 + \left(\frac{1000}{18}\right)}$$

$$= \frac{1}{56.56}$$

Both mole fraction and molality are temperature independent concentration terms.

On dilution, volume of solution increases and hence, given moles will be there in more volume of solution and hence molar concentration decreases.

(52) Answer : (2)

Solution:

$\Rightarrow 0.1 \text{ mol of AB}_3 \text{ weighs} = 11 \text{ g}$

$$\therefore 1 \text{ mole of AB}_3 \text{ weighs} = \frac{11}{0.1} = 110 \text{ g (molar mass)}$$

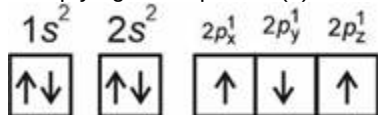
$\Rightarrow 0.5 \text{ mol of A}_2\text{B weighs} = 35 \text{ g}$

$$\therefore 1 \text{ mole of A}_2\text{B weighs} = \frac{35}{0.5} = 70 \text{ g (molar mass)}$$

$$\text{A} + 3\text{B} = 110 \dots \text{(i)}$$

$$2\text{A} + \text{B} = 70 \dots \text{(ii)}$$

Multiplying 2 in equation (1) and subtracting equation (2) from it



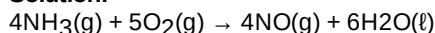
$$\Rightarrow B = \frac{150}{5} = 30$$

$$\text{A} + 3\text{B} = \text{A} + 90 = 110$$

$$\therefore \text{A} = 20$$

(53) Answer : (4)

Solution:



As per reaction stoichiometry 4 mol 1 of ammonia reacts with 5 mol of O_2

So, 1 mol of ammonia reacts with $\frac{5}{4} = 1.25 \text{ mol of O}_2 \text{ (required)}$

$$\text{Available moles of O}_2 = \frac{64}{32} = 2 \text{ mol}$$

Hence, O_2 is excess reagent and NH_3 is limiting reagent. Hence all NH_3 will be consumed

Left mol of O_2 (excess reagent) $= 2 - 1.25 = 0.75 \text{ mol}$

$\Rightarrow 4 \text{ mol of NH}_3 \text{ produces } 4 \text{ mol of NO and } 6 \text{ mol of H}_2\text{O}$

$\therefore 1 \text{ mol of NH}_3 \text{ produces } 1 \text{ mol of NO and } 1.5 \text{ mol of H}_2\text{O}$

(54) Answer : (3)

Solution:

Zeros preceding to first non-zero digit are not significant hence 0.0082 has 2 significant figures zeros between two non-zero digits are significant. 1.004 has 4 significant figures.

Zeros at the end or right of a number are significant provided they are on the right side of decimal point 1.00×10^2 has 3 significant figures.

Counting number of objects have infinite significant figures.

(55) Answer : (1)
Solution:

a. 22400 mL of O_2 at STP $\equiv 2N_A$ oxygen atoms

$$\therefore 2.24 \text{ mL of } O_2 \text{ at STP} \equiv \frac{2N_A}{22400} \times 2.24$$

$$= \frac{2}{10000} \times 6.022 \times 10^{23} \Rightarrow 1.204 \times 10^{20} \text{ atoms}$$

$$\text{b. } 1.4 \text{ g of CO} \equiv \frac{1.4}{28} \times N_A \times 2 \text{ atoms}$$

$$= 0.1 \times 6.022 \times 10^{23} = 6.022 \times 10^{22} \text{ atoms}$$

$$\text{c. } 6.022 \times 10^{23} \text{ molecules of } CH_4$$

$$\equiv 6.022 \times 5 \times 10^{23}$$

$$= 3.0110 \times 10^{24} \text{ atom}$$

$$\text{d. } 1 \text{ mol of } H_2SO_4 \equiv N_A \times 7 \Rightarrow 6.022 \times 10^{23} \times 7$$

$$= 4.21 \times 10^{24}$$

(56) Answer : (2)
Solution:

98% of H_2SO_4 by mass means 98 g H_2SO_4 in its 100 g sample where 2g water is present

$$\text{moles of solute} = \frac{98}{98} = 1 \text{ mol}$$

$$\text{Density of sample} = 1.2 \text{ g mL}^{-1}$$

$$\therefore 100 \text{ g sample has volume} = \frac{1}{1.2} \times 100 \text{ mL}$$

$$= 83.33 \text{ mL}$$

$$\text{Molarity of solution} = \frac{1 \times 1000}{83.33} M$$

Using molarity equation

$$M_1V_1 = M_2V_2$$

$$\frac{1000}{83.33} \times V_1 = 0.1 \times 1$$

$$V_1 = \frac{0.1 \times 83.33}{1000} = 0.00833 L$$

$$= 8.33 \text{ mL}$$

(57) Answer : (4)
Solution:

· Protons and neutrons are present in nuclei and called nucleons

· Number of protons and atomic number of isotopes of elements are same

· Mass of electron = $9.1 \times 10^{-31} \text{ kg}$

Mass of proton = $1.67 \times 10^{-27} \text{ kg}$

· Constituents of cathode ray are electrons

(58) Answer : (3)
Solution:

For a given value of n ,

$$\ell = 0 \text{ to } (n - 1)$$

and for a given value of λ ,

$$m_\ell = -\ell \text{ to } +\ell$$

$$\text{while } m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}$$

Hence, set (ii) ($n = 3$ and $\ell = 3$), set (iii) ($s = +1$) and set (iv) ($m_s = 0$) are not possible.

(59) Answer : (4)
Solution:

	Orbital	Radial node ($n - \ell - 1$)
a.	4s	$4 - 0 - 1 = 3$

b.	3p	$3 - 1 - 1 = 1$
c.	5d	$5 - 2 - 1 = 2$
d.	4f	$4 - 3 - 1 = 0$

(60) Answer : (2)

Solution:

$$\text{Moles of CO}_2 = \frac{100 \times 10^{-3} \text{ g}}{44} \text{ mol} - x = \frac{6.02 \times 10^{20}}{6.02 \times 10^{23}}$$

$$= 0.00227 - x = 0.001$$

$$x = 0.00227 - 0.001 \Rightarrow 0.00127 \text{ mol}$$

$$\therefore \text{Number of molecules} = 0.00127 \times 6.02 \times 10^{23}$$

$$= 7.66 \times 10^{20} \text{ molecules}$$

(61) Answer : (1)

Solution:

Let the mass of urea dissolved in 1 L water = mass of acetic acid dissolved in 1 L water = x g

$$\therefore \text{moles of urea} = \frac{x}{60}$$

$$\text{and moles of acetic acid} = \frac{x}{60}$$

$\therefore$ Molarity of urea = Molarity of acetic acid = $\frac{x}{60} M$ for same moles of solute and solvent in both solutions, mole fraction remain same. Molarity (1 molar concentration) is temperature dependent.

(62) Answer : (3)

Solution:

$$n\lambda = 2\pi r$$

$$\text{For } 2^{\text{nd}} \text{ Bohr orbit } r_2 = a_0 n^2 = 52.9 \times 2^2$$

$$\Rightarrow 211.6 \text{ pm}$$

$$\therefore \lambda = \frac{2\pi \times 211.6}{2} = 211.6 \pi$$

(63) Answer : (4)

Solution:

Bohr's theory could not explain splitting of spectral lines in electric or magnetic field. Splitting of light in the presence of magnetic field is called Zeeman effect while the splitting in electric field is called Stark effect.

(64) Answer : (3)

Solution:

$$2s \Rightarrow \sqrt{1(1+1)} \hbar = \sqrt{0(0+1)} \hbar = 0$$

$$3p \Rightarrow \sqrt{1(1+1)} \hbar = \sqrt{2} \hbar$$

$$4d \Rightarrow \sqrt{2(2+1)} \hbar = \sqrt{6} \hbar$$

$$5f \Rightarrow \sqrt{3(3+1)} \hbar = \sqrt{12} \hbar$$

(65) Answer : (1)

Solution:

Given set of quantum numbers represent an orbital of 4d subshell and an orbital can have maximum of 2 electrons.

(66) Answer : (2)

Solution:

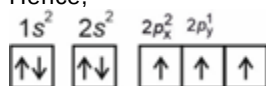
For a p orbital $\ell = 1$ So, possible m_ℓ values = -1, 0, +1

To designate an atomic orbital only 3 quantum numbers n, ℓ and m_ℓ are needed

d_{z^2} orbital has electron density along z-axis

In the same subshell, first orbitals get singly occupied then pairing occurs as per Hund's rule and hence, all the 3 electrons entering to p subshell should have same spin

Hence,



(67) Answer : (4)

Solution:

In disproportionation reaction an element undergoes oxidation and reduction simultaneously. Element in highest or lowest oxidation state cannot disproportionate.

(68) Answer : (3)

Solution:

Species with $10e^- \Rightarrow Na^+, Mg^{2+}, O^{2-}, F^-, Al^{3+}, Si^{4+}$

Species with $18e^- \Rightarrow Ca^+, K^+, Cl^-, Ar$

Species with $14e^- \Rightarrow CO, N_2, O_2^{2+}, CN^-$

(69) Answer : (3)

Solution:

Oxide Nature

Cl_2O_7 Acidic

CO_2 Acidic

Al_2O_3 Amphoteric

BaO Basic

(70) Answer : (2)

Solution:

Lawrencium – 103

Seaborgium – 106

Dubnium – 105

Nobelium – 102

(71) Answer : (3)

Solution:

Electronic configuration of Gallium:

$Ga = [Ar] 3d^{10} 4s^2 4p^1$

(72) Answer : (3)

Solution:

$$r_n = 52.9 \frac{n^2}{z} pm$$

$$\text{Hence, } \frac{(r_2)_{Li^{2+}}}{(r_1)_{He^+}} = \frac{52.9 \times \frac{4}{3}}{52.9 \times \frac{1}{2}} = \frac{8}{3}$$

(73) Answer : (4)

Solution:

Due to stable electronic configuration, noble gases are inert and their electron gain enthalpy values are positive.

If $\Delta_{eg}H$ for Na^+ is x eV then for Na atom $\Delta_f H_1$ will be $-x$ not for Na^+ ion.

(74) Answer : (4)

Solution:

As the nuclear charge increases, the value of ionization energy increases. Also removal of electron from a half filled subshell is also more energetic process.

Hence, ionisation energy of $[Ne]3s^2 3p^3$ is the highest among the given species.

(75) Answer : (1)

Solution:

The correct order of electronegativity is $Si < C < N < O$

(76) Answer : (3)

Solution:

As O^- already has an added electron then the addition of next electron leads to interelectronic repulsion and the process will be endothermic.

(77) Answer : (3)

Solution:

Element	Atomic number	Group
Cd	48	12
Hg	80	12
Al	13	13
Ga	31	13
K	19	1
Sr	38	2
P	15	15
As	33	15

(78) Answer : (2)

Solution:

Chalcogens (p-block elements) are group 16 elements and considered as representative elements.

Element with atomic number more than that of uranium (92) are trans-uranium elements.

So, all actinoids are not trans uranium.

Alkali metal is more electropositive as compared to alkaline earth metal of same period.

In Halogen family,

F₂ – gas

Cl₂ – gas

Br₂ – Liquid

I₂ – Solid

(79) Answer : (3)

Solution:

Given, $\Delta x = \Delta p$

$$ds \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\text{Or } \Delta p^2 = \frac{h}{4\pi}$$

$$(m\Delta v) = \sqrt{\frac{h}{4\pi}} \Rightarrow \sqrt{\frac{h}{4 \times \pi \times m^2}} = \sqrt{\frac{6 \times 10^{-34}}{4 \times \pi \times (9 \times 10^{-31})^2}}$$

$$= \sqrt{\frac{1 \times 10^{28}}{54\pi}}$$

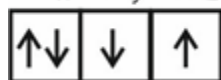
(80) Answer : (3)

Solution:

ψ does not have physical meaning but ψ^2 does.

Two electrons in same orbital can be distinguished by spin quantum number only.

$3p_x \quad 3p_y \quad 3p_z$



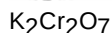
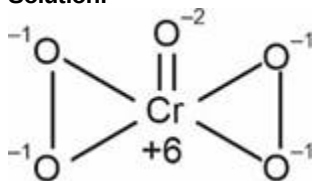
For the two electrons present in $3p_x$ orbital the n , ℓ and m_ℓ values are same

$$n = 3$$

$$\ell = 1$$

$$m_\ell = +1, 0, -1$$

(81) Answer : (2)

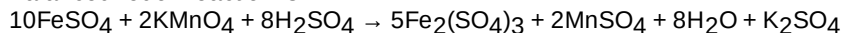
Solution:


$$2(+1) + 2x + 7(-2) \Rightarrow x = +6$$

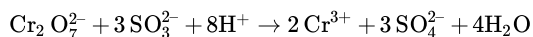
(82) Answer : (1)

Solution:

Balanced redox reaction is



(83) Answer : (4)

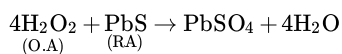
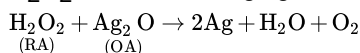
Solution:


3 moles of sulphite ions require 1 mole of acidic $\text{Cr}_2\text{O}_7^{2-}$ ions for oxidation

(84) Answer : (2)

Solution:

H₂O₂ acts as reducing agent in reaction (a) while oxidising agent in reaction (b)



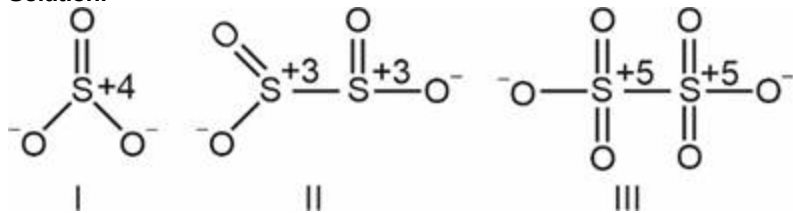
(85) Answer : (3)

Solution:

Oxidizing agent are those species which gains electron i.e. itself gets reduce.
Strongest oxidizing agent have highest value of standard reduction potential.
Therefore Mg^{2+} is the strongest oxidizing agent.

(86) Answer : (1)

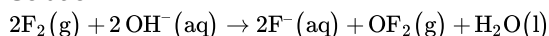
Solution:



Therefore, correct order of oxidation states is $\text{II} < \text{I} < \text{III}$

(87) Answer : (1)

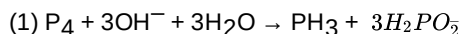
Solution:



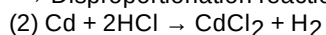
The oxidation state of fluorine changes from 0 to -1 in above reaction.

(88) Answer : (1)

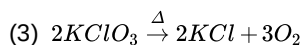
Solution:



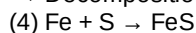
→ Disproportionation reaction



→ Non-metal displacement reaction



→ Decomposition reaction



→ Combination reaction

(89) Answer : (4)

Solution:

$$E_{\text{cell}}^{\circ} = E_{\text{Zn} / \text{Zn}^{2+}}^{\circ} + E_{\text{Fe}^{2+} / \text{Fe}}^{\circ}$$

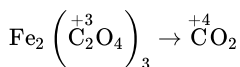
$$= 0.76 \text{ V} - 0.44 \text{ V}$$

$$= 0.32 \text{ V}$$

(90) Answer : (4)

Solution:

When copper rod dipped in silver nitrate solution, colour of solution becomes blue due to formation of Cu^{2+} ions.



Change in oxidation number per carbon atom = $+1$

Total change in oxidation number = $6 \times 1 = 6$

So, number of electrons lost = 6

BOTANY

(91) Answer : (2)

Solution:

Different plant families like Convolvulaceae and Solanaceae are included in the order Polymoniales.

(92) Answer : (1)

Solution:

Actinomorphic flowers are those which can be divided into two equal radial halves in any radial plane passing through the centre, e.g.-Mustard, *Datura*, Chilli.

(93) Answer : (3)

Solution:

Floral formula of a plant does not show the type of inflorescence in that plant.

(94) Answer : (1)

Solution:

Superior ovary – Mustard, China rose

Half-inferior ovary – Plum

Inferior ovary – Guava, Cucumber

Phyllodes – Australian acacia

(95) Answer : (4)

Solution:

In alternate phyllotaxy, a single leaf is present at each node in an alternate fashion, e.g.-China rose, mustard, sunflower.

(96) Answer : (3)

Solution:

Perianth is seen in lily.

(97) Answer : (1)

Solution:

In some plants the axillary buds modify to form tendrils, e.g. watermelon, grapevines, etc.

(98) Answer : (2)

Solution:

The plumule and radical are enclosed in sheaths which are called coleoptile and coleorhiza respectively.

(99) Answer : (1)

Solution:

Endosperm is formed as a result of double fertilization.

In some mature seeds such as castor, it is a food storing tissue.

(100) Answer : (4)

Solution:

Types of aestivation	Example
Valvate	<i>Calotropis</i>
Twisted	Cotton
Imbricate	<i>Cassia</i>
Vexillary	Pea

(101) Answer : (3)

Solution:

The cells proximal to the meristematic region undergo rapid elongation and enlargement and are responsible for the growth of roots in length.

(102) Answer : (3)

Solution:

Types of placentation	Example
Marginal	Pea
Axile	Lemon
Parietal	<i>Argemone</i>
Free-central	<i>Dianthus</i>

(103) Answer : (2)

Solution:

In some plants ovary grows into fruit without fertilisation, such fruits are called parthenocarpic fruits, e.g.- Banana.

(104) Answer : (1)

Solution:

When stamens are united in a single bundle then it is called monoadelphous.

(105) Answer : (1)

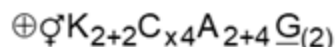
Solution:

When the carpels are fused then they are said to be syncarpous, e.g.- mustard and tomato.

(106) Answer : (1)

Solution:

Mustard belongs to Brassicaceae family and its floral formula is



(107) Answer : (1)

Solution:

The given floral diagram is of makoi plant which belongs to Solanaceae family.

(108) Answer : (3)

Solution:

In the diagram, C represents endocarp which is hard and stony and B represents mesocarp which is inedible.

(109) Answer : (4)

Solution:

Sesbania belongs to the family Fabaceae.

(110) Answer : (2)

Solution:

The first word denoting genus starts with a capital letter, while the specific epithet starts with a small letter.

(111) Answer : (3)

Solution:

Reproduction is shown by all living organisms except a few which are sterile, like mule, worker bees etc. Reproduction can be regarded as characteristics of living organisms but it is not their exclusive defining characteristics.

(112) Answer : (2)

Solution:

Both sieve tube and companion cells are ontogenetically related and hence they are called sister cells

The companion cells help in maintaining the pressure gradient in the sieve tubes.

(113) Answer : (1)

Solution:

Large, empty, colourless cells of the adaxial epidermis along the veins of grass leaves are called bulliform cells.

(114) Answer : (3)

Solution:

Standardisation of scientific name for animals and plants are done by ICZN and ICBN respectively.

(115) Answer : (3)

Solution:

Sclereids are present in pulp of fruits like guava, pear and sapota etc. The gritty texture of these fruits is due to presence of hard cells or sclereids in their pulp.

(116) Answer : (3)

Solution:

Collenchyma is absent in monocotyledonous plants.

(117) Answer : (2)

Solution:

In monocot leaf, mesophyll is not differentiated into palisade and spongy parenchyma.

(118) Answer : (3)

Solution:

The guard cells in the leaves of grasses are dumb-bell shaped.

(119) Answer : (1)

Solution:

Interfascicular cambium is secondary in origin.

(120) Answer : (4)

Solution:

In dicot stem, water containing cavities in vascular bundles are absent, hypodermis is collenchymatous and vascular bundles are conjoint open type.

(121) Answer : (3)

Solution:

Trichomes are usually multicellular and they may be unicellular also.

(122) Answer : (1)

Solution:

Intercalary meristems are found in grasses where they help to regenerate the parts removed by the grazing herbivores.

The cells of secondary meristems divide in radial direction to increase the girth of the plant.

(123) Answer : (2)

Solution:

Both monocot stem and dicot leaf shows bundle sheath around their vascular bundles.

(124) Answer : (1)

Solution:

In cymose inflorescence, the flowers are borne in a basipetal order.

(125) Answer : (2)

Solution:

Sweet pea is an ornamental plant that belongs to Fabaceae family.

(126) Answer : (4)

Solution:

Zygomorphic flowers with marginal placentation are found in the members of Fabaceae family.

(127) Answer : (2)

Solution:

The outer wall of guard cells are thin while the inner wall are highly thickened. Hypodermis is absent in roots. All tissues on the inner side of the endodermis constitute stele.

(128) Answer : (3)

Solution:

Endodermis in roots is characterised by the presence of a special thickened band in their wall called casparian strips.

(129) Answer : (3)

Solution:

Semi-lunar patches of sclerenchymatous pericycle is a feature of dicot stem.

(130) Answer : (3)

Solution:

Cymose inflorescence is seen in *Solanum*.

(131) Answer : (1)

Solution:

The main function of stem is spreading out branches bearing leaves, flowers and fruits.

(132) Answer : (4)

Solution:

Tracheids are found in all categories of vascular plants, i.e., pteridophytes, gymnosperms and angiosperms.

(133) Answer : (2)

Solution:

Systematics takes into account evolutionary relationships between organisms.

(134) Answer : (2)

Solution:

Epiblema gives rise to root hairs. Cuticle is absent in roots. Albuminous cells are present in gymnosperms.

(135) Answer : (2)

Solution:

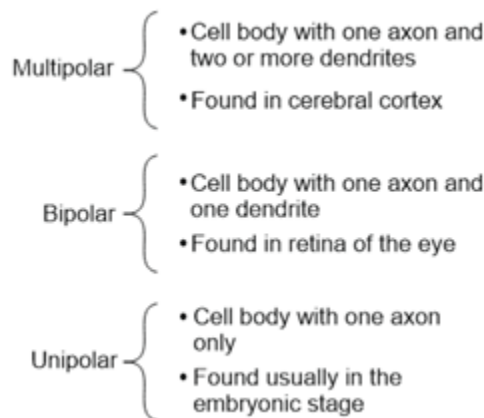
In monocotyledonous plants, the primary root is short-lived and is generally replaced by a number of fine fibrous roots from the base of stem, e.g.-wheat.

ZOOLOGY

(136) Answer : (1)

Solution:

Based on the number of axon and dendrites, the neurons are divided into three types –



(137) Answer : (4)

Solution:

During depolarization of neuronal membrane, the rise in the stimulus – induced permeability to Na^+ is extremely short-lived. It is quickly followed by a rise in permeability to K^+ . Within a fraction of second, K^+ diffuses outside the membrane and restores the resting potential of the membrane at the site of excitation. Influx of Na^+ leads to depolarization of neuronal membrane.

(138) Answer : (2)

Solution:

The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions. It is a part of hindbrain as well as the brain stem. The medulla oblongata passes out through the foramen magnum and continues into spinal cord.

(139) Answer : (3)

Solution:

The cell body (perikaryon) and dendrites of neurons contain certain granular bodies called Nissl's granules. Nissl's granules are made up of rough endoplasmic reticulum and ribosomes; they help in protein synthesis as well.

(140) Answer : (1)

Solution:

Hypothalamus lies at the base of the thalamus. The hypothalamus contains a number of centres which control body temperature, urge for eating and drinking.

(141) Answer : (1)

Solution:

Fibres of the tracts are covered with the myelin sheath, which constitute the inner part of the cerebral hemispheres. They give an opaque white appearance to the layer and hence, is called the white matter.

(142) Answer : (3)

Solution:

During depolarization, rapid influx of Na^+ leads to reversal of the polarity at the site of stimulus applied, i.e., the outer surface of the membrane becomes negatively charged and the inner side becomes positively charged.

At the resting stage of an axon, the membrane potential is not at the equilibrium potential for K^+ due to slight permeability to other ions like sodium, resulting in a resting potential of -70 mV .

(143) Answer : (4)

Solution:

If receptors on the post-synaptic membrane are removed, the neurotransmitter released from the pre-synaptic neuron will not be able to bind to the post-synaptic membrane. Thus, action potential will not develop in the post-synaptic neuron.

(144) Answer : (2)

Solution:

In humans, the brain is covered by cranial meninges consisting of an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer called pia mater.

(145) Answer : (3)

Solution:

The neural organisation is very simple in lower invertebrates. For example, in *Hydra*, it is composed of a network of neurons. The neural system is better organised in insects, where a brain is present along with a number of ganglia and neural tissues.

(146) Answer : (1)

Solution:

In a chemical synapse, action potential travels along the neurons from the dendrites through the cell body and down the axon towards the axon terminals. When the action potential reaches the synaptic knob, it causes the release of neurotransmitters into the synaptic cleft.

(147) Answer : (4)

Solution:

The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called the limbic lobe or the limbic system. Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions, autonomic responses, olfaction and motivation.

(148) Answer : (2)

Solution:

The gaps between two adjacent myelin sheaths are called nodes of Ranvier. Nodes of Ranvier are absent in the unmyelinated axons.

(149) Answer : (3)

Solution:

Oxytocin is formed in the hypothalamus but released by the posterior pituitary. The cerebral cortex contains motor areas, sensory areas and large regions that are neither clearly sensory nor motor in function. These regions, known as the association areas, are responsible for complex functions like intersensory associations, memory and communication.

(150) Answer : (2)

Solution:

Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts. Hormones which interact with intracellular receptors (e.g., steroid hormones, iodothyronines, etc.) mostly regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome. Amino acid derived hormones and protein hormones are water-soluble.

(151) Answer : (4)

Solution:

Aldosterone is the main mineralocorticoid in our body. Hypothalamic hormones are proteinaceous in nature.

(152) Answer : (4)

Solution:

Adrenaline and noradrenaline are rapidly secreted in response to stress of any kind and are called emergency hormones or hormones of fight or flight. These hormones increase alertness, pupillary dilation, piloerection (raising of hair), sweating, etc. Both hormones increase the heart beat, the strength of heart contractions and the rate of respiration. Catecholamines stimulate the breakdown of glycogen resulting in an increased concentration of glucose in blood. In addition, they also stimulate the breakdown of lipids and proteins.

(153) Answer : (4)

Solution:

Ovaries produce relaxin which is a protein hormone.

Goitre can be caused by both hyposecretion (Simple goitre) and hypersecretion (Graves' disease) of thyroxine hormone.

(154) Answer : (2)

Solution:

Thyroxine is an iodothyronine and glucocorticoids (cortisol) are steroidal in chemical nature. Glucocorticoids produce anti-inflammatory reactions and suppress the immune responses. Both, thyroxine and glucocorticoids, control the metabolism of carbohydrates, proteins and fats.

(155) Answer : (4)

Solution:

Protein hormones react with receptors on the surface of the cell, and the sequence of events that results in hormone action is relatively rapid. Steroidal hormones (testosterone, cortisol, etc.) typically react with receptor sites inside a cell. Since their method of action involves synthesis of proteins, their effect is observed at relatively slower rate.

(156) Answer : (2)

Solution:

The adrenal cortex can be divided into three layers. They are:

1. Zona reticularis - Inner layer
2. Zona fasciculata - Middle layer and
3. Zona glomerulosa - Outer layer

Aldosterone is released by the outermost layer of the adrenal cortex, called zona glomerulosa.

Aldosterone is the major mineralocorticoid. It regulates homeostasis of sodium and potassium ions, and helps adjust blood pressure and blood volume.

(157) Answer : (4)

Solution:

Progesterone and estrogen: These are steroid hormones produced mainly by the ovaries (and placenta during pregnancy). They help in mammary gland development during pregnancy, preparing the mammary glands for lactation.

Oxytocin: It is a peptide hormone released by the posterior pituitary. It plays a key role in the milk ejection reflex (let-down reflex) by stimulating the contraction of mammary glands.

(158) Answer : (1)**Solution:**

The hormones originating in the hypothalamic neurons, pass through axons and are released from their nerve endings. These hormones reach the pituitary gland through a portal circulatory system and regulate the functions of the anterior pituitary.

The posterior pituitary is under the direct neural regulation of the hypothalamus.

(159) Answer : (1)**Solution:**

Catecholamines act *via* membrane-bound receptors and generate second messengers. They are released by the adrenal medulla. Cortisol acts *via* intracellular receptors. Pineal gland secretes melatonin; and it is present on the dorsal side of forebrain. Thymus secretes thymosins; and it is present in the thoracic region. The adrenal glands are located in the upper abdomen.

(160) Answer : (4)**Solution:**

In males, androgens stimulate muscular growth, growth of facial and axillary hair, aggressiveness, low pitch of voice, *etc.* It also plays a major stimulatory role in the process of spermatogenesis.

The Leydig cells or interstitial cells, which are present in the intertubular spaces produce a group of hormones called androgens mainly testosterone.

(161) Answer : (4)**Solution:**

Pituitary (Prolactin and MSH), thyroid, adrenal, pancreas, parathyroid, thymus (thymosin) and gonads are the organised endocrine bodies in our body.

(162) Answer : (1)**Solution:**

The hypothalamus possesses the centres for control of body temperature. The thyroid gland plays an important role in regulation of BMR, thus, influences body temperature.

The pineal gland also influences body temperature. Thymus gland secretes thymosins which play a major role in the differentiation of T-lymphocytes.

(163) Answer : (1)**Solution:**

Gastrin – Stimulates secretion of gastric gland

Renin converts angiotensinogen into angiotensin I

ANF is a potent vasodilator

GIP (gastric inhibitory peptide) – Inhibits gastric secretion and motility

(164) Answer : (3)**Solution:**

Some specialised cells in our body like macrophages and leucocytes in blood exhibit amoeboid movement. Cytoskeletal elements like microfilaments are also involved in amoeboid movement.

Porifers have flagellated choanocytes.

(165) Answer : (2)**Solution:**

Smooth muscle fibres have tapering ends (fusiform) and do not show striations. Cell junctions hold them together. They are unbranched and uninucleated.

Cardiac muscle fibres are branched. Skeletal muscle fibres and cardiac muscle fibres show striations.

Skeletal muscle fibres are multinucleated.

(166) Answer : (4)**Solution:**

During the muscle contraction, the cross-bridge formed, pulls the attached actin filaments towards the centre of 'A' band.

The 'Z' line attached to these actins also gets pulled inwards, thereby causing shortening of the sarcomere (distance between successive 'Z' lines). The 'I' bands and 'H' zone gets reduced, whereas 'A' bands retain their length.

(167) Answer : (3)**Solution:**

At the point of fusion of ilium, ischium and pubis is a cavity called acetabulum to which the thigh bone articulates.

Glenoid cavity and acromion are associated with the pectoral girdle.

The two halves of the pelvic girdle meet ventrally to form the pubic symphysis containing fibrous cartilage.

(168) Answer : (3)

Solution:

A motor neuron transmits an electrical signal to its terminal at the neuro-muscular junction. This triggers the release of the neurotransmitter acetylcholine at neuro-muscular junction. Acetylcholine binds to receptors present on the sarcolemma, leading to depolarization and the initiation of an action potential in the muscle fibres.

The action potential travels along the sarcolemma reaching the sarcoplasmic reticulum. This triggers the release of stored calcium ions into the sarcoplasm.

Ca^{2+} binds to troponin. This binding causes a conformational change in another protein, tropomyosin, exposing binding sites on actin.

Myosin heads of the thick filaments bind to these exposed sites on actin, forming cross-bridges. This pulls the attached actin filaments towards the centre of 'A' band.

This sliding action shortens the sarcomere, leading to muscle contraction.

ATP binds to myosin heads, causing them to detach from actin.

The hydrolysis of ATP re-energizes the myosin heads, allowing them to reattach to a new position on the actin filament, continuing the cycle as long as Ca^{2+} and ATP are available.

(169) Answer : (2)

Solution:

'I' band contains only actin.

'A' band contains both actin and myosin.

(170) Answer : (1)

Solution:

Myoglobin content is high in some of the muscles which gives them a reddish appearance. Such muscles are called the red muscle fibres. These muscles also contain plenty of mitochondria which can utilise the large amount of oxygen stored in them for ATP production. These muscles, therefore, can also be called aerobic muscles.

(171) Answer : (4)

Solution:

Knee joint is a type of hinge joint. Saddle joint is present between carpal and metacarpal of thumb.

(172) Answer : (2)

Solution:

Muscle fibre is the anatomical unit of muscle. Each muscle fibre has many parallelly arranged myofibrils. Each myofibril contains many serially arranged units called sarcomere which are the functional units.

Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles.

(173) Answer : (3)

Solution:

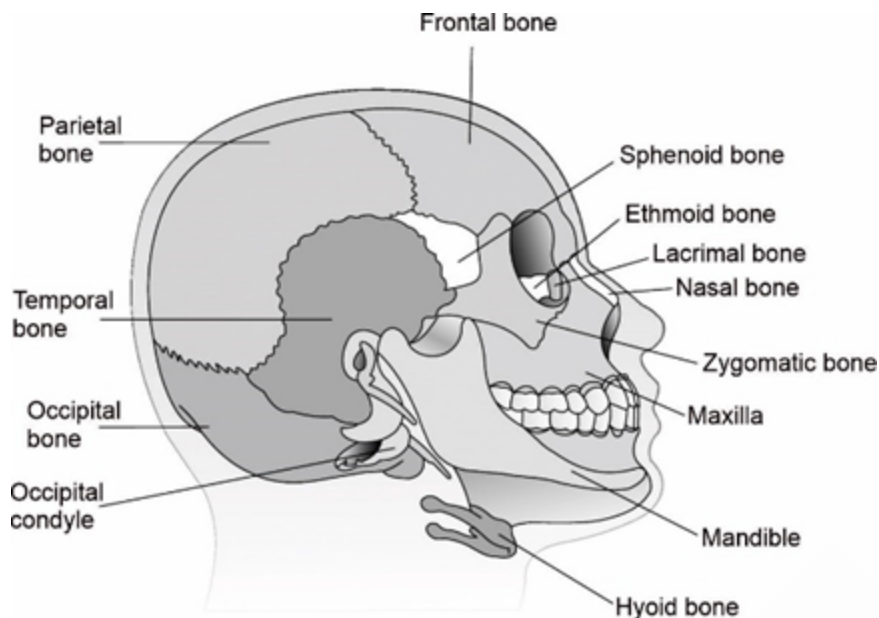
Myasthenia gravis: Auto-immune disorder affecting neuro-muscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

Osteoporosis: Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.

Gout: Inflammation of joints due to accumulation of uric acid crystals.

(174) Answer : (2)

Solution:



Parietal and temporal bones are paired cranial bones.

(175) Answer : (3)

Solution:

Thoracic vertebrae – 12

True ribs – 7 pairs (14)

Tarsals in both hind limbs – 14

Pectoral girdle – 4

(176) Answer : (2)

Solution:

The bones of the limbs along with their girdles constitute the appendicular skeleton.

The skull, vertebral column, sternum and ribs constitute the axial skeleton.

(177) Answer : (1)

Solution:

Sternum is a part of the axial skeleton. Clavicle is a part of appendicular skeleton. It is also called the collar bone. Fibula does not articulate directly with patella.

(178) Answer : (3)

Solution:

The globular head of meromyosin is an active ATPase enzyme and has binding sites for ATP and active sites for actin.

(179) Answer : (3)

Solution:

Femur is the longest bone in the human body.

Sternum is a flat bone on the ventral midline of thorax.

A cup-shaped bone, called patella, covers the knee ventrally (knee cap).

(180) Answer : (3)

Solution:

Progressive degeneration of skeletal muscle mostly due to genetic disorder is called muscular dystrophy.

Tetany is caused due to low calcium in blood and myasthenia gravis is an auto-immune disorder.